

기상기후빅데이터

Weather and Climate Big Data

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Today's Topics

- Introduction to Report Writing
- Review of Related Works
- Report Submission

Why Academic Paper Format?

- Clear logic: **Introduction → Method → Results → Discussion**
- Arguments must be supported by **evidence and citations**
- Weather and climate data require scientific interpretation
- Builds skills for future thesis or research writing
- Not just a summary – it's **analytical thinking training**

Report Writing

- **Overall Structure**
 - Abstract
 - Introduction
 - Methods
 - Results and Discussion
 - Conclusion
 - Reference

Report Writing

- **Overall Structure**

- **Abstract:** a concise summary of the entire project, including its goals, data, methods, and key outcomes.
- **Introduction:** an overview of the topic that provides background, motivation, and a brief review of related work.
- **Methods and Data:** a detailed description of the data used and the analytical or computational methods applied.
- **Results and Discussion:** a presentation of the main findings, along with interpretation, significance, and limitations.
- **Conclusion:** a closing section that highlights the main insights and potential future directions.
- **Reference:** a list of all sources, datasets, and works cited, formatted consistently.

Report Writing

- **Overall Structure**

- **Abstract:** a concise summary of the planned project, including main objectives, potential data sources or methods, and expected outcomes.
- **Introduction:** background and motivation for the proposed project
- **Methods and Data:** description of the data you plan to use (or are considering), how it will be collected or prepared, and the methods or tools you intend to apply.
- **Results and Discussion:** (To be completed later) Optionally, you may include expected results or hypotheses at this stage.
- **Conclusion:** (To be completed later)
- **Reference:** a list of any papers, datasets, or sources you've referred to while preparing the proposal.

Review of Related Works

- **Why Review Existing Studies?**
 - Understand what has already been done on your topic
 - Get inspiration for your own research direction
 - Learn about commonly used data, methods, and variables
 - Support your data interpretation with previous findings

Review of Related Works

- Where to find related papers?
 - Use platforms like **Google Scholar**, **ResearchGate**, and **open-access journals** (e.g., Copernicus publications)
 - Focus on **review articles** or papers from **high-impact journals** to understand key research trends
 - Check the **reference lists** of relevant papers to trace foundational or related studies

Review of Related Works

- How to use prior studies
 - Review prior studies for their data, methods, and conclusions
 - Clearly explain how your project is different or builds upon past work
 - Use previous research to guide your analysis and interpretation
 - Focus on studies from the last 10 years
 - Read high-quality or review papers to understand strong research examples

Clonal dynamics of native haematopoiesis

Jianlong Sun^{1,2,3}, Azucena Ramos¹, Brad Chapman⁴, Jonathan B. Johnnidis⁵, Linda Le¹, Yu-Jui Ho⁶, Allon Klein⁷, Oliver Hofmann⁴ & Fernando D. Camargo^{1,2,3}

It is currently thought that life-long blood cell production is driven by the action of a small number of multipotent haematopoietic stem cells. Evidence supporting this view has been largely acquired through the use of functional assays involving transplantation. However, whether these mechanisms also govern native non-transplant haematopoiesis is entirely unclear. Here we have established a novel experimental model in mice where cells can be uniquely and genetically labelled *in situ* to address this question. Using this approach, we have performed longitudinal analyses of clonal dynamics in adult mice that reveal unprecedented features of native haematopoiesis. In contrast to what occurs following transplantation, steady-state blood production is maintained by the successive recruitment of thousands of clones, each with a minimal contribution to mature progeny. Our results demonstrate that a large number of long-lived progenitors, rather than classically defined haematopoietic stem cells, are the main drivers of steady-state haematopoiesis during most of adulthood. Our results also have implications for understanding the cellular origin of haematopoietic disease.

implication of results (they even say "implications"!) →

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Ecology drives a global network of gene exchange connecting the human microbiome

Chris S. Smillie^{1*}, Mark B. Smith^{2*}, Jonathan Friedman¹, Otto X. Cordero³, Lawrence A. David⁴ & Eric J. Alm^{3,5,6}

general background → Horizontal gene transfer (HGT), the acquisition of genetic material from non-parental lineages, is known to be important in bacterial evolution^{1,2}. In particular, HGT provides rapid access to genetic innovations, allowing traits such as virulence³, antibiotic resistance⁴ and xenobiotic metabolism⁵ to spread through the human microbiome. Recent anecdotal studies providing snapshots of active gene flow on the human body have highlighted the need to determine the frequency of such recent transfers and the forces that govern these events^{4,5}. Here we report the discovery and characterization of a vast, human-associated network of gene exchange, large enough to directly compare the principal forces shaping HGT. We show that this network of 10,770 unique, recently transferred (more than 99% nucleotide identity) genes found in 2,235 full bacterial genomes, is shaped principally by ecology rather than geography or phylogeny, with most gene exchange occurring between isolates from ecologically similar, but geographically separated, environments. For example, we observe 25-fold more HGT between human-associated bacteria than among ecologically diverse non-human isolates ($P = 3.0 \times 10^{-270}$). We show that within the human microbiome this ecological architecture continues across multiple spatial scales, functional classes and ecological niches with transfer further enriched among bacteria that inhabit the same body site, have the same oxygen tolerance or have the same ability to cause disease. This structure offers a window into the molecular traits that define ecological niches, insight that we use to uncover sources of antibiotic resistance and identify genes associated with the pathology of meningitis and other diseases.

specific background →

knowledge gap →

here we show... →

results with key, concrete values →

meaning of results →

acquired from close relatives, because these genes have greater compatibility with native molecular machinery^{15,16}.

Geography might provide an alternative structure to HGT by restricting dispersal, as suggested by the geographically organized distribution of *Vibrio cholera* integrins¹⁷ and NDM-1 antibiotic resistance genes¹⁸.

A third possibility is that ecological similarity shapes networks of gene exchange by selecting for the transfer and proliferation of adaptive traits or by increasing physical interactions between community members. Reports of enriched levels of HGT between hyperthermophiles¹⁹ and spatially segregated exchange among *Shewanella* isolates²⁰ offer suggestive glimpses of such an ecological structure. However, it has been difficult to determine whether ecology has a broader function in HGT because of the limited availability of genomes from similar environments and because most previous work has ignored the distinction between recent transfers and ancient events. The inclusion of transfers from millions or billions of years in the past can obscure ecological structure, because historical niches may not reflect modern environmental associations.

To explore the effects of phylogeny, geography and ecology on HGT we use an evolutionary-rate heuristic to identify recent transfers between thousands of microbial genomes. Our heuristic finds blocks of nearly identical DNA (more than 500 nucleotides, more than 99% identity) in distantly related genomes (less than 97% 16S rRNA similarity). HGT is the best explanation for these observations because the highly conserved 16S gene evolves about 25-fold more slowly than protein-coding synonymous sites²¹. As a result, vertically inherited orthologues in such divergent genomes are nearly saturated with mutations at

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The human body is a complex biological network comprising ten

중간고사 기획안 제출 안내

- 4월 27일 일요일 자정 까지 이루리캠퍼스 “과제”에 제출
- 작성 범위: 초록(Abstract), 서론(Introduction), 방법(계획) 부분까지 작성
- 최종 제출은 반드시 **PDF 형식**으로 저장하여 제출
- 기상·기후 데이터를 활용. 단, 환경 및 지구시스템 관련 데이터(예: 대기, 수문, 해양 등)도 자유롭게 선택 가능
- 기말에는 실제 분석 결과물 제출이 필수
 - 분석과 결과 도출이 가능한 범위 내에서 주제를 선정할 것
- ChatGPT 등 AI 도구 사용은 허용되나, 어디에 어떻게 사용했는지 반드시 문서 내에 명시